UNIT 7 · STANDARD 6.EE.8

Graph Inequalities

Lesson 7-5

Name: Date: Class:

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A horizontal line with marks at 0, 1, 2, 3, ... showing where values fall

Number line

A line with numbers in order. You use it to show answers.

$x > 5$: open circle at 5 means 5 itself is NOT a solution, but 5.1, 6, 7, ... are

Open circle

An empty circle showing a number is NOT included.

$x \geq 5$: closed circle at 5 means 5 IS a solution, and so are 6, 7, 8, ...

Closed circle

A filled-in circle showing a number IS included.

For $x > 3$, the solution set includes 3.1, 4, 5, 10, 100 — any number greater than 3

Solution set

All the numbers that make the inequality true.

$x \geq 3$ is an inequality; it is graphed as a closed circle on 3 with an arrow to the right.

Inequality

A math sentence comparing two amounts with <, >, ≤, or ≥.

For $x > 5$, the boundary point is 5, shown with an open circle.

Boundary point

The number on the number line where the solution set starts.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can graph the solutions of an inequality on a number line.

1️⃣ Notice

Detective Okafor is building a timeline.

2️⃣ Key idea

I can graph the solutions of an inequality on a number line.

3️⃣ Words I'll use

Number line

Open circle

Closed circle

Solution set

Inequality

Boundary point

🤔 Think About It

- What does 'no earlier than 2:00 PM' mean about the time?
- Should 2:00 PM itself be included?
- Which direction on the number line would you shade?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Which graph represents $x < 5$?

A. Open circle at 5, shaded left

B. Closed circle at 5, shaded left

C. Open circle at 5, shaded right

D. Closed circle at 5, shaded right

Step 1 $x < 5$ means 5 is NOT included (open circle) and values less than 5 are shaded (left).

✅ **Answer:** A. Open circle at 5, shaded left

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which graph represents $x \geq 3$?

A. Closed circle at 3, shaded right

B. Open circle at 3, shaded right

C. Closed circle at 3, shaded left

D. Open circle at 3, shaded left

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which graph represents $x > 8$?

A. Open circle at 8, shaded right

B. Closed circle at 8, shaded right

C. Open circle at 8, shaded left

D. Closed circle at 8, shaded left

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Okafor knows the suspect arrived 'no earlier than 2:00 PM,' so $t \geq 2$. On the timeline, should the mark at 2 be an open or closed circle, and which way should she shade?

Level 1 support

Start here: Because 2:00 itself counts, I use a closed circle at 2.

💡 **Need a hint?**

Sentence starters (optional)

I use a ____ circle at 2 because 2:00 ____ included.

Uso un círculo ____ en 2 porque las 2:00 ____ incluidas.

I shade ____ because the times can be ____.

Sombreo ____ porque las horas pueden ser ____.

Word bank: number line | open circle | closed circle | solution set

Listen for: *Students choose a closed circle at 2 (because 'no earlier than' includes 2) and shade to the right since later times satisfy $t \geq 2$.*

Level 2

How would the timeline look different if the clue were $t > 2$ instead of $t \geq 2$? Justify the change in the circle.

- For $t > 2$ I would use a ____ circle because ____.
- The boundary value 2 would ____ in that case.

EXPLORE

When you graphed $x > 4$ and $x \le 7$, why did you use an open circle on one and a closed circle on the other?

Level 1 support

Start here: Open circles exclude the boundary; closed circles include it.

💡 **Need a hint?**

Sentence starters (optional)

I used a ___ circle because the symbol ___ means the boundary is ___.

Usé un círculo ___ porque el símbolo ___ significa que el límite ___.

I shaded ___ because the solution set includes ___.

Sombreé ___ porque el conjunto solución incluye ___.

Word bank: open circle | closed circle | number line | solution set

Listen for: *Students connect > and < to open circles (boundary not included) and ≥ and ≤ to closed circles, and shade toward the values in the solution set.*

Level 2

Why does the graph of $x > 4$ represent infinitely many solutions even though it starts at one point? Explain using the solution set.

- The graph shows infinitely many solutions because ___.
- Every point in the shaded part is ___.

CONNECT

When would graphing an inequality show a real-life limit better than a single number?

Level 1 support

Start here: A graph shows every value that fits a rule, not just one.

💡 **Need a hint?**

Sentence starters (optional)

A graph shows a real limit like ____ because ____.

Una gráfica muestra un límite real como ____ porque ____.

The ____ circle shows whether the limit value ____ allowed.

El círculo ____ muestra si el valor límite ____ permitido.

Word bank: number line | open circle | closed circle | solution set

Listen for: *Students name a real limit (minimum age, max weight, earliest time) and explain the circle type shows whether the boundary value is allowed.*

Level 2

Critique: 'An open circle and a closed circle graph the same solution set.' Use $x > 5$ and $x \geq 5$ to show why this is wrong.

- This is wrong because $x \geq 5$ includes ____ but $x > 5$ ____.
- The two solution sets differ by ____.

Write About the Math

The Writing Revolution

I can explain my graph using the words number line, open circle, closed circle, and solution set.

1. Kernel Sentence

subject + verb

Model: Boundary point is the number on the number line where the solution set starts.
Punto límite es el número en la recta numérica donde comienza el conjunto solución.

Write a kernel sentence about boundary point. Use a subject and a verb.
Escribe una oración base sobre punto límite. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Boundary point matters in math
Punto límite importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Boundary point matters in math because ____.** *Punto límite importa en matemáticas porque ____.*
but *pero*	**Boundary point matters in math, but ____.** *Punto límite importa en matemáticas, pero ____.*
so *entonces*	**Boundary point matters in math, so ____.** *Punto límite importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about boundary point.
Di un hecho verdadero sobre boundary point.

Boundary point ____.

Question
Pregunta

Ask a question about boundary point.
Haz una pregunta sobre boundary point.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about boundary point.
Muestra entusiasmo sobre boundary point.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with boundary point.
Dile a un compañero qué hacer con boundary point.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I use a ____ circle at 2 because 2:00 ____ included.
Uso un círculo ____ en 2 porque las 2:00 ____ incluidas.

I shade ____ because the times can be ____.
Sombreo ____ porque las horas pueden ser ____.

I used a ____ circle because the symbol ____ means the boundary is ____.
Usé un círculo ____ porque el símbolo ____ significa que el límite ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A graph shows a closed circle at -2 shaded to the right. Which inequality is it?

A. $x \geq -2$

B. $x > -2$

C. $x \leq -2$

D. $x < -2$

Show your work:

2. Which value is a solution to $x < 4$?

A. 3

B. 4

C. 5

D. 6

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A theme park has two rides. Ride A requires height h > 48 inches. Ride B requires height h ≥ 48 inches. A child is exactly 48 inches tall. Which ride can they go on? Explain using number line graphs for both inequalities.

Sentence starter: For Ride A (h > 48), the graph has a ____ circle at 48. Since 48 > 48 is ____, the child ____ ride A. For Ride B (h ≥ 48), the graph has a ____ circle at 48. Since 48 ≥ 48 is ____, the child ____ ride B.

Show your work:

Reflect — Exit Ticket

How would you graph the inequality x ≥ 9?

A. Closed circle at 9, shaded right

B. Open circle at 9, shaded right

C. Closed circle at 9, shaded left

D. Open circle at 9, shaded left

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. Closed circle at 3, shaded right *— $x \geq 3$ means 3 IS included (closed circle) and values greater than or equal to 3 are shaded (right).*
2. **You Try (notes):** A. Open circle at 8, shaded right *— $x > 8$ means 8 is NOT included (open circle) and values greater than 8 are shaded to the right.*
3. **Try It 1:** A. $x \geq -2$ *— Closed circle means 'or equal to'; shaded right means greater: $x \geq -2$.*
4. **Try It 2:** A. 3 *— Only 3 is less than 4.*
5. **Exit Ticket:** A. Closed circle at 9, shaded right *— $x \geq 9$ means 9 is included (closed circle) and values are 9 or greater (shaded right).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Boundary point is the number on the number line where the solution set starts.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).